

Reading Report: Solvable Model for Discrete Time Crystal Enforced by Nonsymmorphic Dynamical Symmetry

Reproduction notes for the Hu–Fu–Li–Shen DTC project

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Abstract

This report summarizes the physical background, central model, symmetry mechanism, special parameters, and numerical reproduction targets of Zi-Ang Hu, Bo Fu, Xiao Li, and Shun-Qing Shen, “Solvable model for discrete time crystal enforced by nonsymmorphic dynamical symmetry,” *Physical Review Research* **5**, L032024 (2023). The focus is on understanding how nonsymmorphic dynamical symmetry enforces a period-doubled response in a solvable two-level Floquet model.

1 Background

A time crystal is a dynamical phase or regime where time-translation symmetry is broken in the observed motion. For a continuously time-independent equilibrium Hamiltonian, spontaneous breaking of continuous time translation is forbidden. Periodically driven Floquet systems provide a different setting: the Hamiltonian is invariant under discrete time translation by one drive period T , while observables may respond with a larger period, usually $2T$. This is the defining signature of a discrete time crystal (DTC).

Earlier DTC proposals often relied on disorder, many-body localization (MBL), or prethermal regimes to avoid featureless Floquet heating. MBL is important because it can preserve local memory in disordered interacting systems, allowing a subharmonic response to remain rigid against weak imperfections. However, strict MBL depends strongly on disorder, dimensionality, and finite-size assumptions. Prethermal DTCs relax some of these requirements by producing an exponentially long-lived regime before eventual heating.

The Hu–Fu–Li–Shen paper takes a different starting point. It asks whether nonsymmorphic dynamical symmetry can enforce period extension already at the single-spin level. Interactions then stabilize the observable subharmonic response by suppressing fast off-diagonal oscillations, but the basic period-doubling mechanism originates from symmetry and topology of the instantaneous eigenstates.

2 Two-Level Model

The central Hamiltonian is

$$H_0(t) = \frac{1}{2} \hbar \Omega \sin(\omega t) \sigma_x + \hbar \Omega \sin^2\left(\frac{\omega t}{2}\right) \sigma_y, \quad (1)$$

where $\sigma_{x,y,z}$ are Pauli matrices. The drive period is

$$T = \frac{2\pi}{\omega}. \quad (2)$$

The paper defines the dimensionless parameter

$$\alpha = \frac{8\Omega}{\omega}. \quad (3)$$

For the reproduction work, the dimensionless convention is

$$\hbar = 1, \quad \omega = 1, \quad \Omega = \alpha/8, \quad T = 2\pi.$$

The instantaneous eigenstates can be written, up to the sign convention used in the paper, as

$$\phi_+(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\omega t/2} \\ 1 \end{pmatrix}, \quad \phi_-(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} -e^{-i\omega t/2} \\ 1 \end{pmatrix}. \quad (4)$$

At $t = 0$, $\phi_+(0)$ is the $+x$ -polarized spin state.

3 Nonsymmorphic Dynamical Symmetry

The model has a dynamical glide symmetry. The symmetry operator has the same period T as the Hamiltonian, but its eigenvalues and eigenstates interchange after one period and return only after two periods. In this sense, the instantaneous eigenstate bundle has a Mobius twist.

The physical consequence is a mismatch between two periods. The Hamiltonian itself is T -periodic, while the irreducible representation carried by the instantaneous eigenstates has period $2T$. This mismatch enforces a subharmonic response without relying on an ordinary avoided-crossing adiabatic picture.

4 Exact Solution and Special Parameters

In the instantaneous eigenbasis, the time-dependent Schrodinger equation reduces to a second-order equation whose solution is written in terms of confluent Heun functions. The one-period probability for an initial $\phi_+(0)$ state to remain in $\phi_+(0)$ is controlled by the Heun-function value at $x = \pi/2$.

There is a discrete sequence α_n where this one-period return probability vanishes. At these values, the state is mapped after one period to the partner instantaneous state up to a phase, and after two periods it returns to the initial state with a Berry phase of π . This realizes the DTC response in the solvable two-level model.

The paper lists the following approximate values:

Table 1: Approximate special values α_n from the paper.

n	1	2	3	4	5	6	7
α_n	4.21	10.73	17.11	23.44	29.75	36.07	42.36
n	8	9	10	11	12	13	14
α_n	48.66	54.95	61.24	67.46	73.76	80.07	86.34

For large n , the asymptotic estimate is

$$\alpha_n \simeq 2\pi n - \frac{\pi}{2}. \quad (5)$$

The noninteracting reproduction focuses on the table value near α_{13} and on the detuned value $\alpha = 81.60$, which is used in the paper's noninteracting Fig. 4(a).

5 Observables and Reproduction Targets

The main noninteracting observable is

$$\langle \sigma_x(t) \rangle = \langle \psi(t) | \sigma_x | \psi(t) \rangle, \quad (6)$$

for the initial $+x$ -polarized state. When α is detuned from α_n , the off-diagonal term in the instantaneous basis contains a rapid dynamical phase factor,

$$\exp \left[-i\alpha \sin^2 \left(\frac{\omega t}{4} \right) \right], \quad (7)$$

which creates rapid oscillations and multiple Fourier peaks in $\langle \sigma_x(t) \rangle$. In the interacting model, the paper argues that interactions suppress these fast oscillations and stabilize a cleaner period-doubled average magnetization.

The first reproduction stage therefore has six concrete targets:

1. implement the two-level Hamiltonian in Eq. (1);
2. use the initial $+x$ -polarized state;
3. evolve over the first 40 drive periods;
4. compute $\langle \sigma_x(t) \rangle$ and its Fourier spectrum for $\alpha = 81.60$;
5. compare a unitary fourth-order Magnus solver with a fourth-order Runge–Kutta solver;
6. verify that the refined value near α_{13} produces nearly perfect two-period stroboscopic behavior.

6 Conclusion

The key lesson of the paper is that a nonsymmorphic dynamical symmetry can impose a Mobius twist on instantaneous eigenstates, making the state evolution subharmonic to the Hamiltonian drive. This provides an analytically controlled route to a DTC-like response. The noninteracting calculation tests the basic symmetry-enforced mechanism, while the interacting extension addresses stabilization and prethermal lifetime.